



Grids of the Future

Increasing residential solar adoption in Singapore

Team SOLAR - SG

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Executive Summary of SOLAR-SG

Problem Statement

Accelerating the adoption of solar PV in the residential households in Singapore & engaging the currently untapped market of solar enthusiast community

About Us

Vision

To pioneer a localised online solar community that boosts solar adoption & interest

Mission

A centralised, all-in-one, social online platform for solar PV data gathering, optimization and resource sharing

Target



\$ 3 M

Forecasted
revenue by
2028



> 500,000

Active user
base by
2028

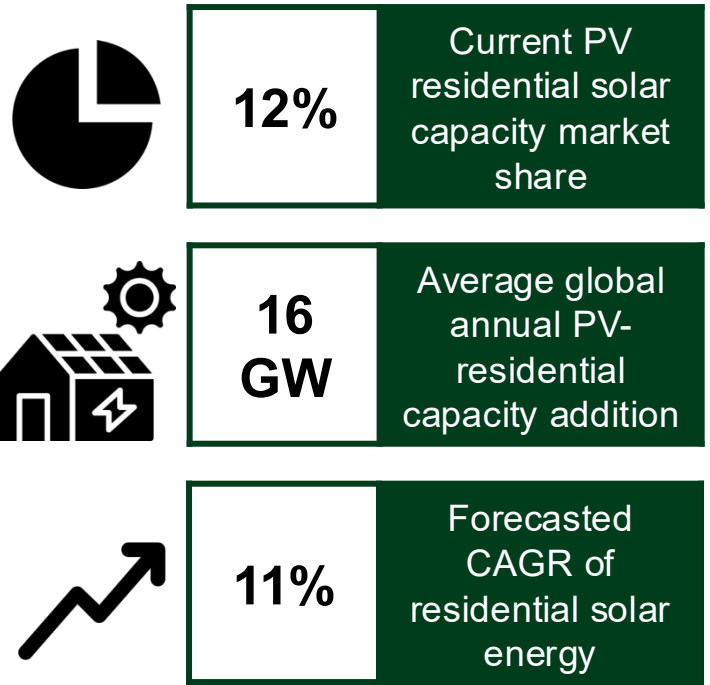
Conclusion

Through our integrated online services, SOLAR-SG establishes a **reliable go-to platform** for **local solar enthusiasts** to explore and, ultimately, adopt solar PV

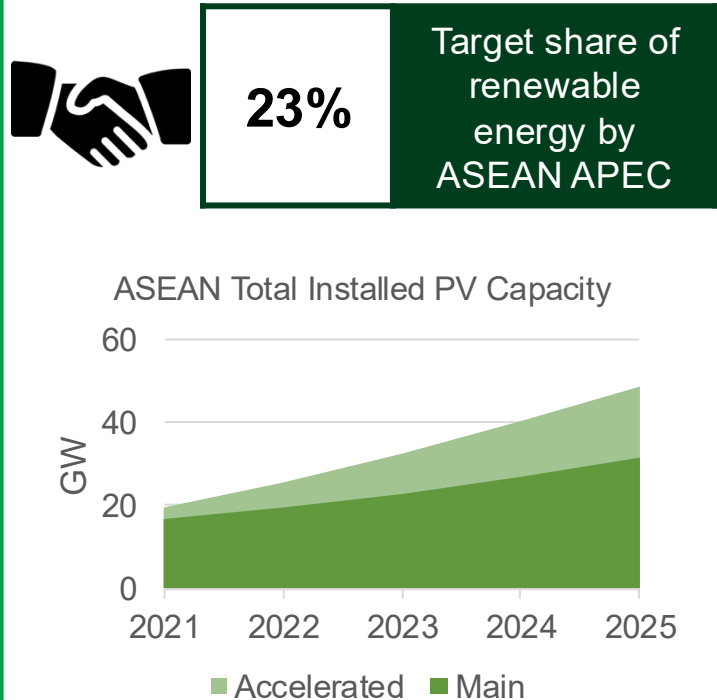
Aligned with the Singapore's Green plan & UNSDG

There is a promising growth in the global, ASEAN & Singapore solar energy market for SOLAR-SG to tap on

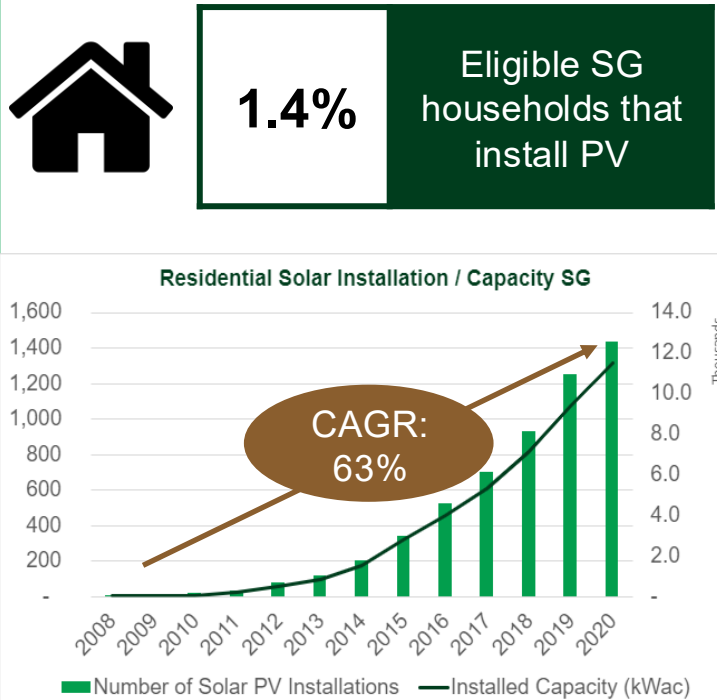
Global solar PV capacity additions is expected to accelerate in 2023-2025



Great opportunity to penetrate the ASEAN market due to improved regulation



Huge untapped residential market opportunity in Singapore



Source: Rodriguez, 2021; IEA, 2020; Mordor Intelligence, 2022; EMA, 2021



Insight

Residential solar PV presents a huge **untapped** but growing market for SOLAR-SG to penetrate

Factors affecting solar energy adoption

1  Inaccurate payback period calculation

2  Perceived benefits of solar energy

3  Social influence

4  Prior product knowledge & experience

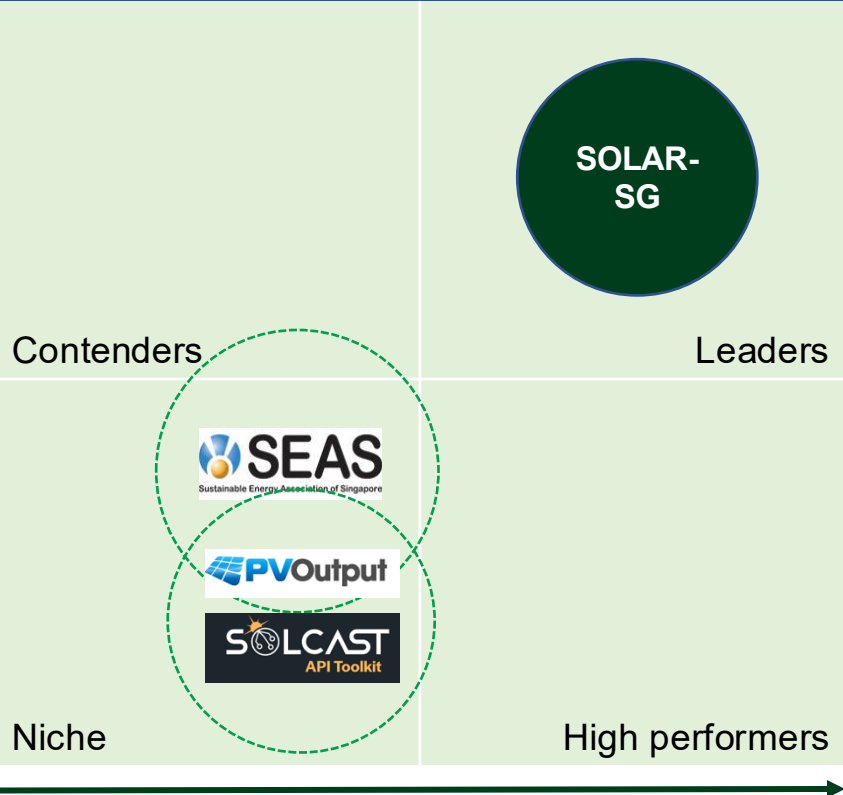


There is a need to

provide information and a change in culture to encourage adoption of solar panels

Competitive landscape

Penetration



Performance

Existing solar PV monitoring solutions are lacking in

 Localised data

 Local online community

 Contextualised features

 Marketing & publicity

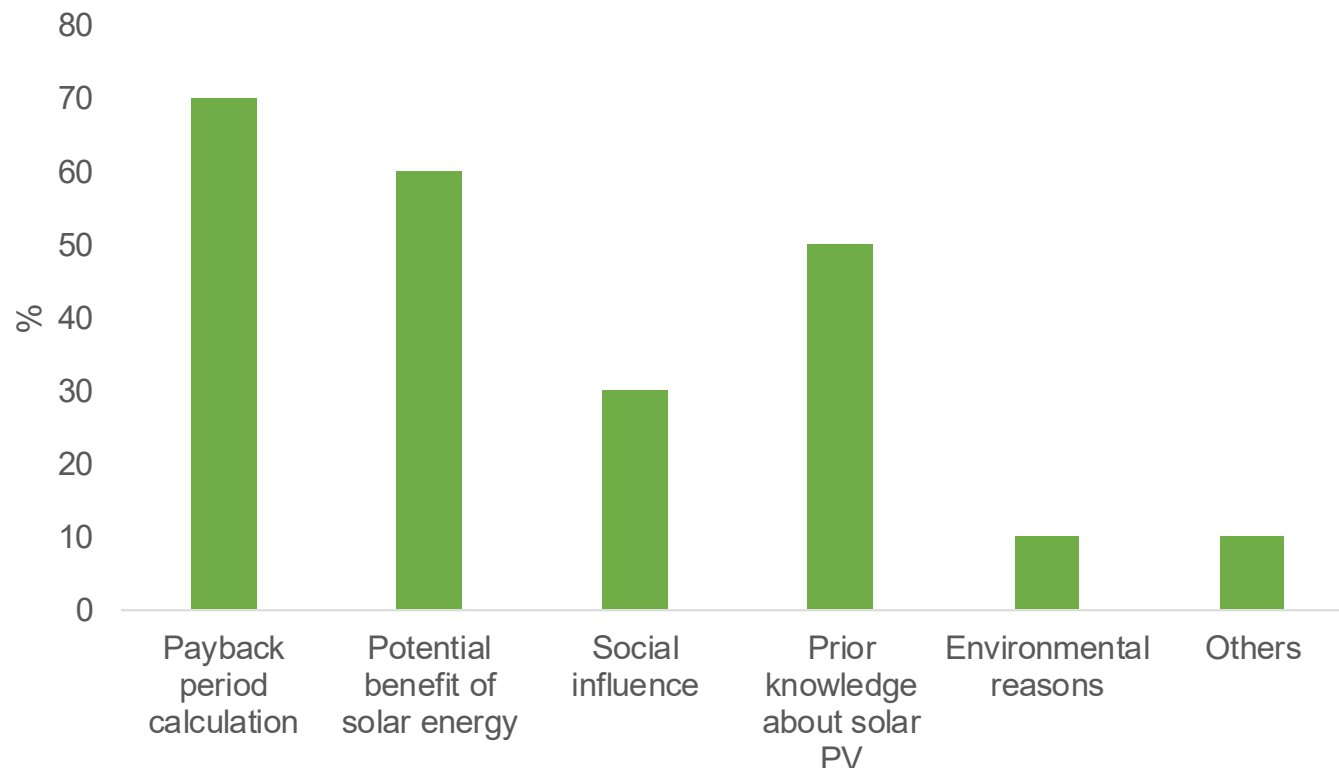
Our market survey indicates significant interest and willingness to adopt solar PV & confirms factors affecting adoption

Customer Profile

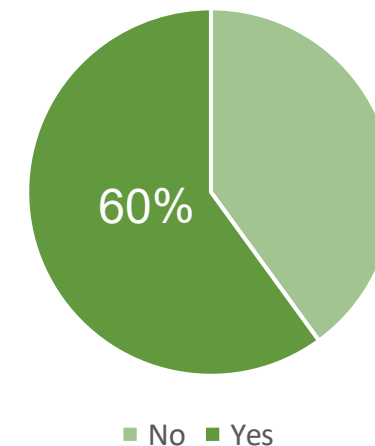
Tech-savvy individual who **wants to learn more** about solar energy or **thinking of installing solar PV** in their residential houses

From our market survey of 200 respondents in Singapore aged 20 to 60

Factors affecting one's decision to install solar PV



Willing to invest in solar PV given enough financial means to do so

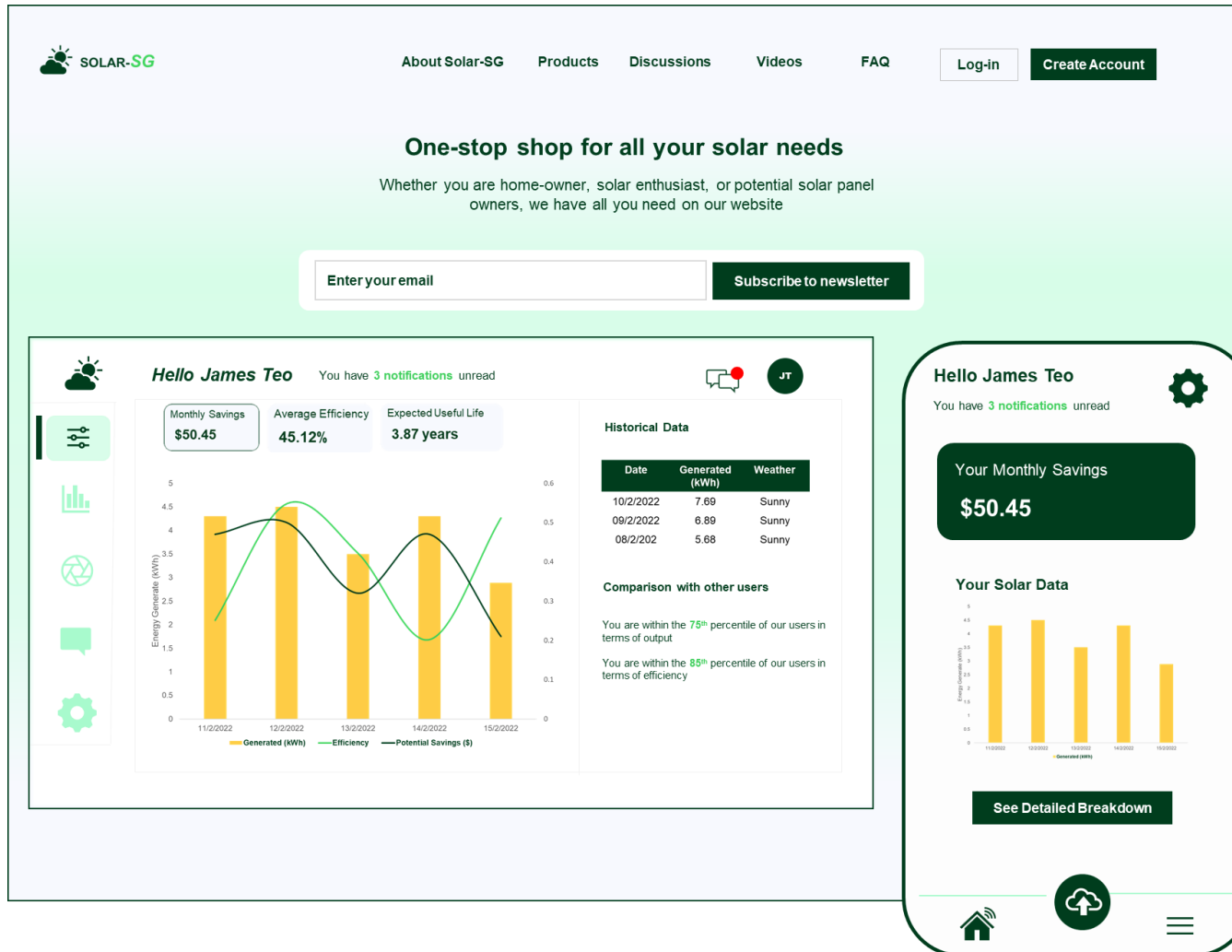


55%

More incentivised given easy all-in-one online/mobile platform as guidance

SOLAR-SG aims to provide a one-stop website for solar panel enthusiasts and the public

Value Proposition: SOLAR-SG is a **consolidated platform** that aims to **create interest and increase adoption** in solar panel by providing useful **information and insights** as well as **encouraging interactions** among users.



Key Features

- Data sharing** – Uploading of solar related data
- Data analytics** – Analysis of user data
- Discussions** – Commenting function, Forums
- Learning** – Videos, Contents provided by other users
- Products** – Online marketplace for vendors to advertise their products

SOLAR-SG also aims to bridge the current gaps in the solar panel market in order to increase adoption

SOLAR-SG aims to increase the adoption of solar panel by **simulating a peer pressure effect** and **lowering the barriers of entry** through offering **learning resources** and valuable information about the **direct benefits to users**.

Current Gaps

1

Lack of **financial** metrics



2

Lack of **customised** analytical tool for personal solar panels



3

Information are **scattered and haphazard**



4

Lack of **established and structured** platform for solar enthusiasts to gather



5

Lack of **historical data** of solar panels to carry out research



SOLAR-SG

The one-stop solution for solar panels in Singapore



SOLAR-SG's value proposition

1

Calculates the direct **cost savings** for users for using solar panel



2

Learning resources, discussions and products available are **consolidated**



3

Personalised data analytics based on user inputs



4

Discussion forums and commenting functions to **encourage interactions** and to **form strong bonds**



5

Users' contributions provide valuable insights about **regional solar usage**





SOLAR-SG

Consumer User Experience

User Retention

"How do I monitor my energy performance?"



User Loyalty

"How do I connect with like-minded people?"



User Awareness

"How do I go green?"



User Onboarding

"How do I tap on Solar Energy?"



Profile:

Has a family
Advanced in career

Expectations:

Name:

James Teo

- Investment opportunity
- Contribute to environment

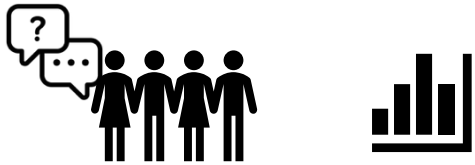


SOLAR-SG

Captures and delivers value through its business model

SOLAR-SG can create revenue through...

1 Sale of analytics reports



Collecting user inputs

Generate market sentiments and technical data

Produce **technical and marketing reports** to be sold to businesses

2 Listing Revenue

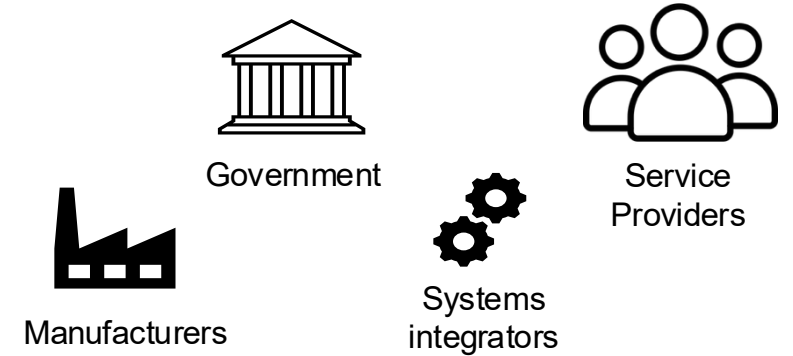


Online marketplace for solar power partners

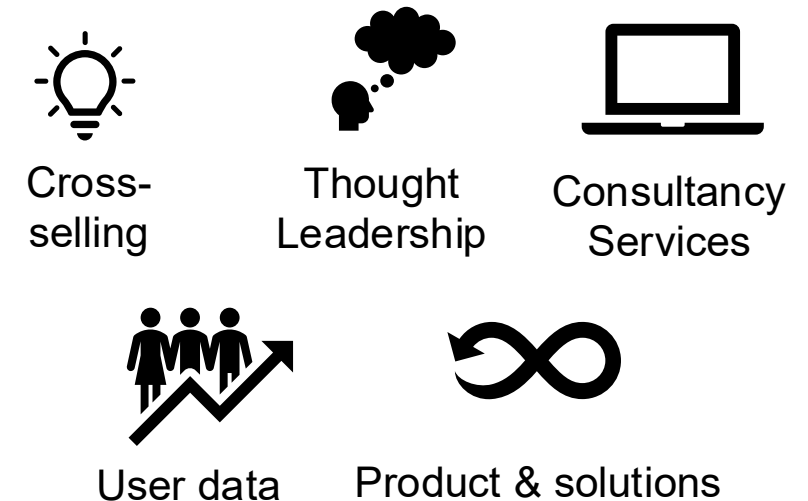


Charging **listing fees** to manufacturers, system integrators and service providers

Delivering value to...

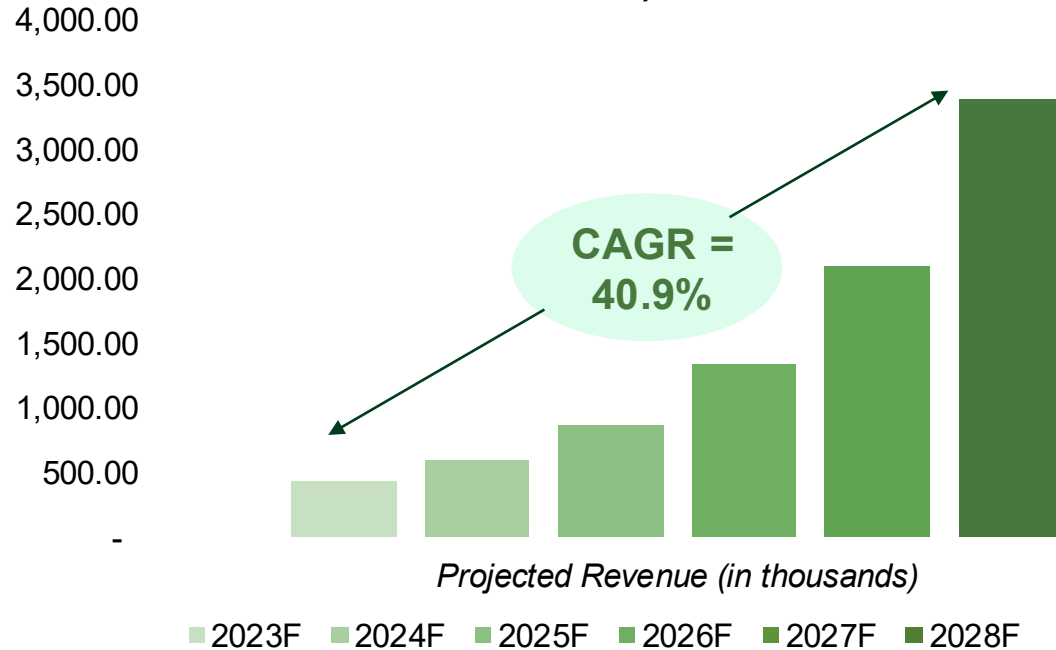


How Schneider benefits





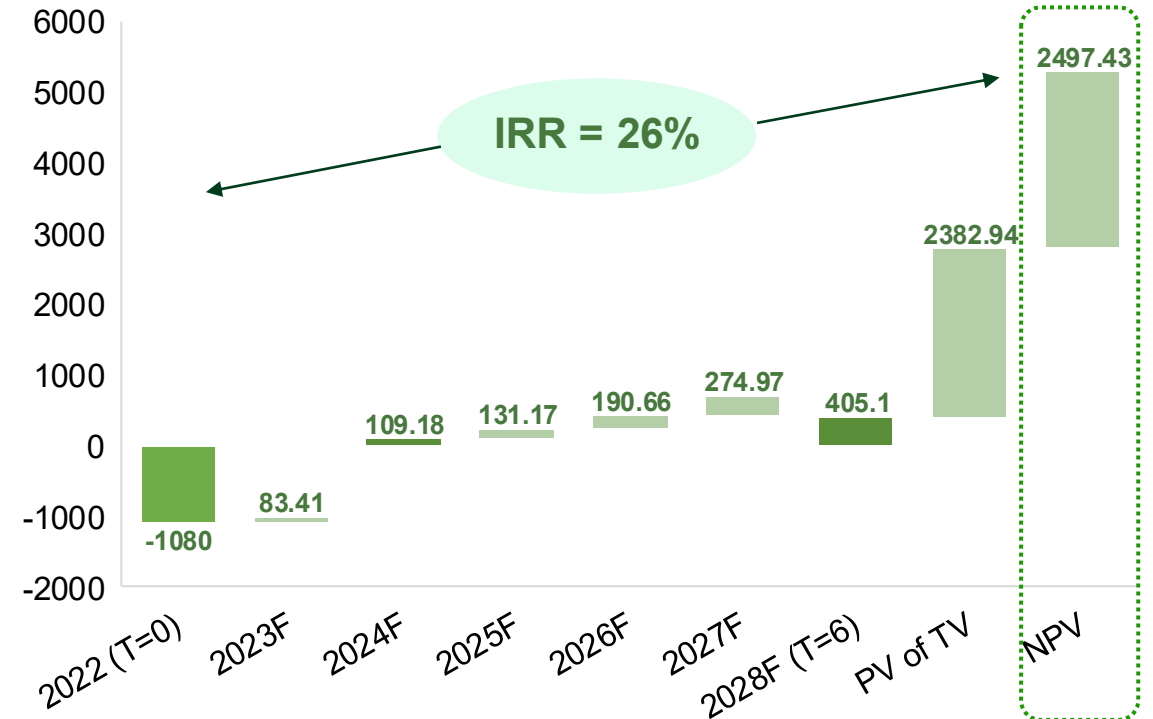
Revenue Forecasts from this project (in thousands SGD)



Forecast of Revenue

- User subscriptions are free of charge
- Revenue generated through listing revenues and sale of user analytics and insights
- Forecasted using a bottoms-up approach based on **number of paid and organic users** as well as the **listing revenues** charged to vendors

Cash Flows and NPV (in thousands SGD)



Basis of calculations

- Initial investment outlay for app and webpage creation = 1.08 million
- Operating expenses include maintenance, administrative and marketing costs – expected to be **proportional with revenue growth**
- Terminal value calculated using **Gordon growth formula**

Promotes greater adoption and efficiency of solar power, driving sustainability

Relating to GHG Protocols and SBTi...

Scope 1



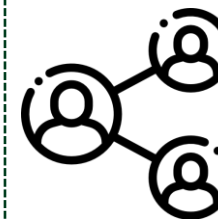
Landed-property owners reduce direct emissions

Scope 2



Apartment dwellers purchase solar from provider

Scope 3



Promotion of solar and greater R&D for panel efficiency

Potential Impacts

25 tonnes

of CO₂ equivalent GHG emissions reduced per household per month

Aim to reduce at least

178,000 tonnes

of CO₂ equivalent emissions by 2026.

Relating to UNSDG goals

7 AFFORDABLE AND CLEAN ENERGY



11 SUSTAINABLE CITIES AND COMMUNITIES



12 RESPONSIBLE CONSUMPTION AND PRODUCTION



13 CLIMATE ACTION



17 PARTNERSHIPS FOR THE GOALS



THANK YOU

Q&A



Users Forecast	2023	2024	2025	2026	2027	2028
Paid	42,500	56,525	79,700	118,753	182,880	290,780
Organic	42,500	53,833	72,290	102,584	150,456	227,833
Total Users	85,000	110,358	151,991	221,337	333,336	518,613
Paid/Organic Ratio	1	1.1	1.1	1.2	1.2	1.3

Paid users growth rate						
Average Growth Rate		38%	46%	54%	59%	64%
Average Churn Rate		-5%	-5%	-5%	-5%	-5%
Net Growth Rate		33%	41%	49%	54%	59%

Revenue per user						
Data Analytics	5.00	5.25	5.62	6.01	6.25	6.50
Revenue per vendor						
Listing Fees	150	165.00	181.50	181.50	181.50	181.50
Vendor Growth						
No. of vendors	50	60	72	86	104	124

Revenue						
Sale of data	425,000	579,381	853,808	1,330,396	2,083,736	3,371,606
Listing Revenue	7,500	9,900	13,068	15,682	18,818	22,582
Total Revenue	432,500	589,281	866,876	1,346,077	2,102,554	3,394,188

- Bottoms-up approach

Target Users (in thousands)

Singapore Population

(20-64 years old) 2565.7

Target Users by 2028

(20%)* 513.14

Reference: Singstat. (n.d.). Population and Population Structure. Retrieved from: <https://www.singstat.gov.sg/find-data/search-by-theme/population/population-and-population-structure/latest-data>

- *We aim to accumulate more than 20% of the 20-64 years old population by 2028



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- We also aim to increase the paid/organic ratio by increasing the amount of marketing programmes
- Average growth rate will increase every year until we reach a constant rate – at that point we will expand into the international market
- We aim to maintain the churn rate at about 5% per year

Initial Investments to create the app

Complex App	900,000.00	Reference: Spd.load(n.d.). <i>How Much Does It Cost to Develop an App in 2022? Cost Breakdown.</i> Retrieved from: https://spdload.com/blog/app-development-cost/
Administrative Costs (10%)	<u>180,000.00</u>	
	<u>1,080,000.00</u>	

Period	2022 0	2023 1	2024 2	2025 3	2026 4	2027 5	2028 6
Investment Outlay	(1,080,000.00)						
PV (OCF)		83,413.93	109,178.12	131,168.19	190,661.51	274,969.60	405,100.26
PV (Terminal Value)							2,382,942.69
PV(CF)	(1,080,000.00)	83,413.93	109,178.12	131,168.19	190,661.51	274,969.60	2,788,042.95

NPV	2,497,434.30
IRR	26%

Required rate of return	22%	<i>Higher than average rate of return due to riskiness of the project</i>
Terminal Growth Rate	5%	





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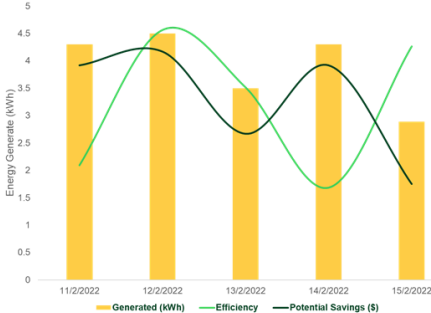
Hello James Teo

You have 3 notifications unread

Monthly Savings
\$50.45

Average Efficiency
45.12%

Expected Useful Life
3.87 years



Historical Data

Date	Generated (kWh)	Weather
10/2/2022	7.69	Sunny
09/2/2022	6.89	Sunny
08/2/2022	5.68	Sunny

Comparison with other users

You are within the 75th percentile of our users in terms of output

You are within the 85th percentile of our users in terms of efficiency







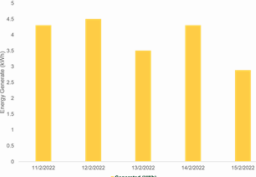


Hello James Teo

You have 3 notifications unread

Your Monthly Savings
\$50.45

Your Solar Data



[See Detailed Breakdown](#)

Users can subscribe to the monthly newsletter to keep themselves updated about developments of solar panel

The app version allows for easy upload of solar panel information

SOLAR-SG also provides a variety of functions to keep users engaged while using the app

Users can see each other's uploads and interact through the comment function

Projected no. of solar PV installations by scenario

	Scenario: No Solar-SG - grow at current rate	Scenario: Implement Solar-SG - grow at optimistic rate
Year	Growth at current rate (CAGR 13%)	Growth at optimistic rate (CAGR 26%)
2021	1,613.86	1,796.73
2022	1,820.09	2,255.92
2023	2,052.68	2,832.47
2024	2,314.98	3,556.38
2025	2,610.80	4,465.29
2026	2,944.43	5,606.50

Key assumptions

- Each household doing panel installation requires 1,182 kWh of electricity per month
- Each solar PV installation reduces 360g CO₂ eq / kWh when benchmarked against natural gas
- Natural gas is assumed to be the household's alternative source of electricity
- Does not account for carbon debt of installing solar panels

Average energy consumption

Terrace house	890 kwh per month
Semi-detached	1182 kwh per month

Source of power	Amount of CO ₂ eq GHG emitted by energy source (g CO ₂ eq/KWh)	CO ₂ eq GHGs emitted per terrace household per month in g
Solar	40	2836800
Natural gas	400	28368000

Carbon emission savings for converting to solar per terrace house per month in kg

25531.2

Emission savings in 2022	11,127,192.25
Emission savings in 2023	19,909,135.06
Emission savings in 2024	31,694,319.60
Emission savings in 2025	47,347,259.83
Emission savings in 2026	67,965,807.16
Total carbon emission savings from project in kg (2022-2026)	178,043,713.89
Total carbon emission savings from project in tonnes (2022-2026)	178,043.71